Elasticity

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Introduction

Classification of materials:

Based on the behavior of the material on the application of external force (called deforming force): 1. Rigid, 2. Elastic and 3. Plastic.

1. **Rigid:** It does not undergo any deformation when external forces act on it. When forces are applied on a rigid body the distance between any two particles of the body will remain unchanged however large the force may be. In actual practice there is no perfect rigid body. But solids can be treated as rigid if the applied force is not enough to deform them.

2. **Elastic:** It undergoes deformation when suitable external force acts on it. The change could be either in shape or size or even both. If the body recovers its original state after removal of the deforming force then it is called an elastic body. Eg., quartz.

   Elasticity is the property of the material of a body by virtue of which it regains its original shape and size after the deforming force is removed.

3. **Plastic:** It does not show any tendency of returning back to its initial state and stays in the changed form even after the removal of external force. Eg., clay.

   Plasticity is the property of the material of a body by virtue of which it fails to regain its original shape and size after the deforming force is removed.

Atomic picture of elasticity:

- Atoms in a solid material are held together by electrostatic force of attraction (intermolecular binding forces).
- Under the applied deforming force the molecules are displaced from their original position.
- When displaced the atoms develop a tendency to come back to their original position because of the intermolecular binding forces.
- The aggregate of the restoration tendency exhibited by all the molecules of the body, manifests as a balancing force or restoring force counteracting the external force.
- When the external forces are removed the intermolecular forces tend to regain the shape and size of the body.

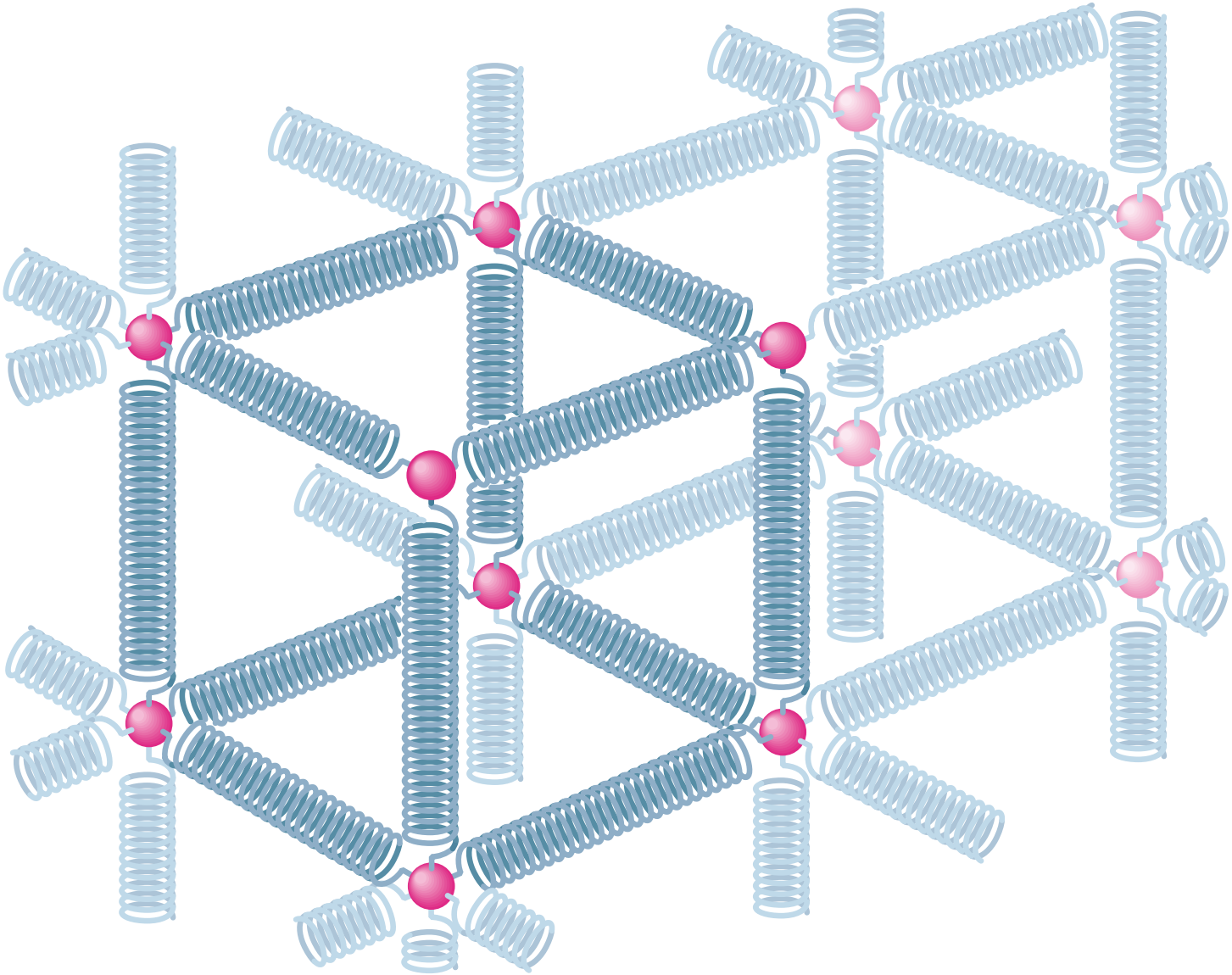


Figure: The atoms of a metallic solid are distributed on a repetitive three dimensional lattice. The springs represent interatomic forces.

Important Terms to be Understood

- **Deforming Force:** The applied external force, which causes, deformation is called deforming force.
- **Restoring force:** The reaction force which is developed in a deformed body is called restoring force.
- **Load:** It is the external forces acting on a body is the load and its effect is to change the form or the dimensions of the body.
- **Strain:** The deformation produced by the external force accompanying a change in dimensions or form of the body is called strain. It is the ratio of change in dimensions of the body to its original dimensions.
- **Stress:** It is the ratio of restoring force to the area of its application. That is, stress $= F/A$. Unit of stress is Nm^{-2}.
  - **Normal Stress:** Restoring force per unit area perpendicular to the surface is called normal stress.
  - **Tangential Stress:** Restoring force parallel to the surface per unit area is called tangential stress.

When calculating stress, the formula is indeed stress = force/area. Typically, the area used in this calculation is the initial area, which is the cross-sectional area before any deformation occurs. This is particularly important in cases like tensile or compressive stress, where the material may deform under load.

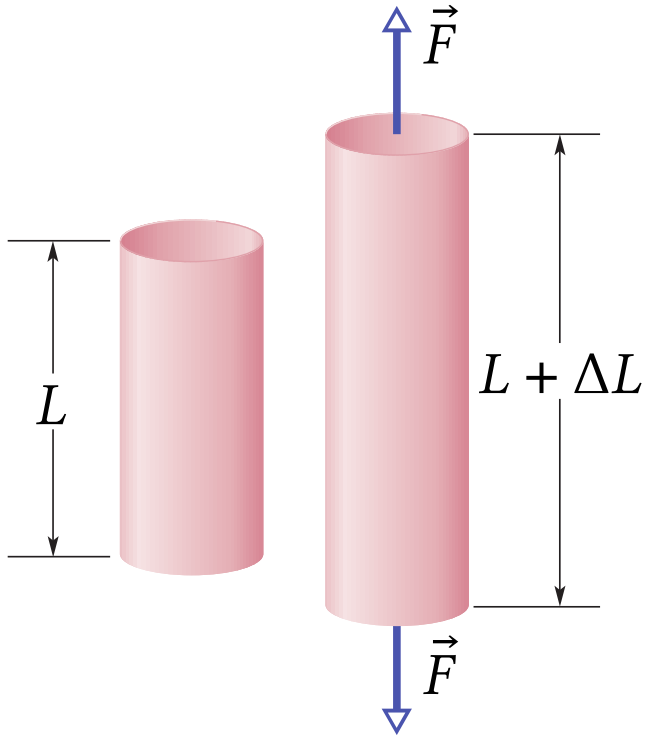
If the material undergoes significant deformation, the final area might be used in specific contexts, like in large deformations or in some advanced analyses, but for most standard calculations, the initial area is the correct choice.

Strain

The deformation produced by the external force accompanying a change in dimensions or form of the body is called strain. It is the ratio of change in dimensions of the body to its original dimensions.

Deformation is of three types, resulting in three types of strains.

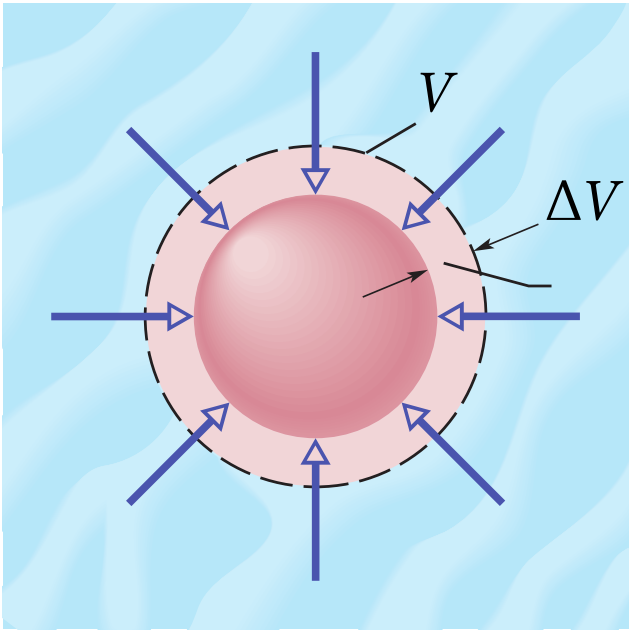
1. **Linear strain or Tensile strain:** If the shape of the body could be approximated to the form of a long wire and if a force is applied at one end along its length keeping the other end fixed, the wire undergoes a change in length.
   If ΔL is the change in length produced for an original length L then,



$$\text{linear strain} = \frac{\text{change in length}}{\text{original length}} = \frac{\Delta L}{L}$$

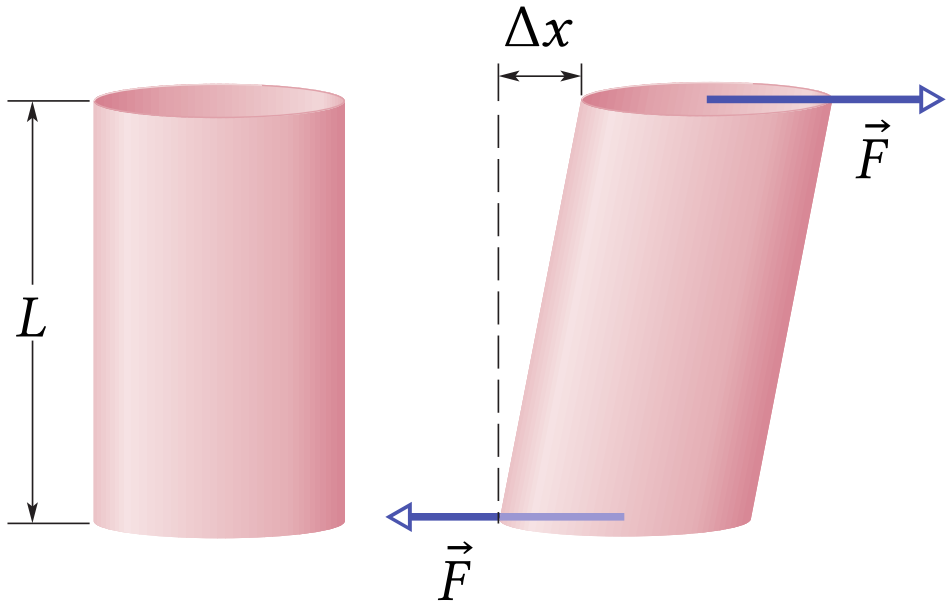
2. **Volume strain or Bulk strain:** If a uniform force is applied all over the surface of a body, the body undergoes a change in its volume (however the shape is retained in case of solid bodies).
If ΔV is the change in volume to an original volume V of the body, then,

$$\text{volume strain} = \frac{\text{change in volume}}{\text{original volume}} = \frac{\Delta V}{V}$$



3. **Shear strain:** If a force is applied tangentially to a free surface of a body with the other surface being fixed, its layers slide one over the other. The body experiences a turning effect and changes its shape. This is called shearing and the angle through which the turning takes place is called shearing angle.

$$\text{shear strain, } \theta = \frac{\Delta x}{L}$$



Hooke's Law

It states that *"the stress is directly proportional to the strain"* (provided strain is small). That is,

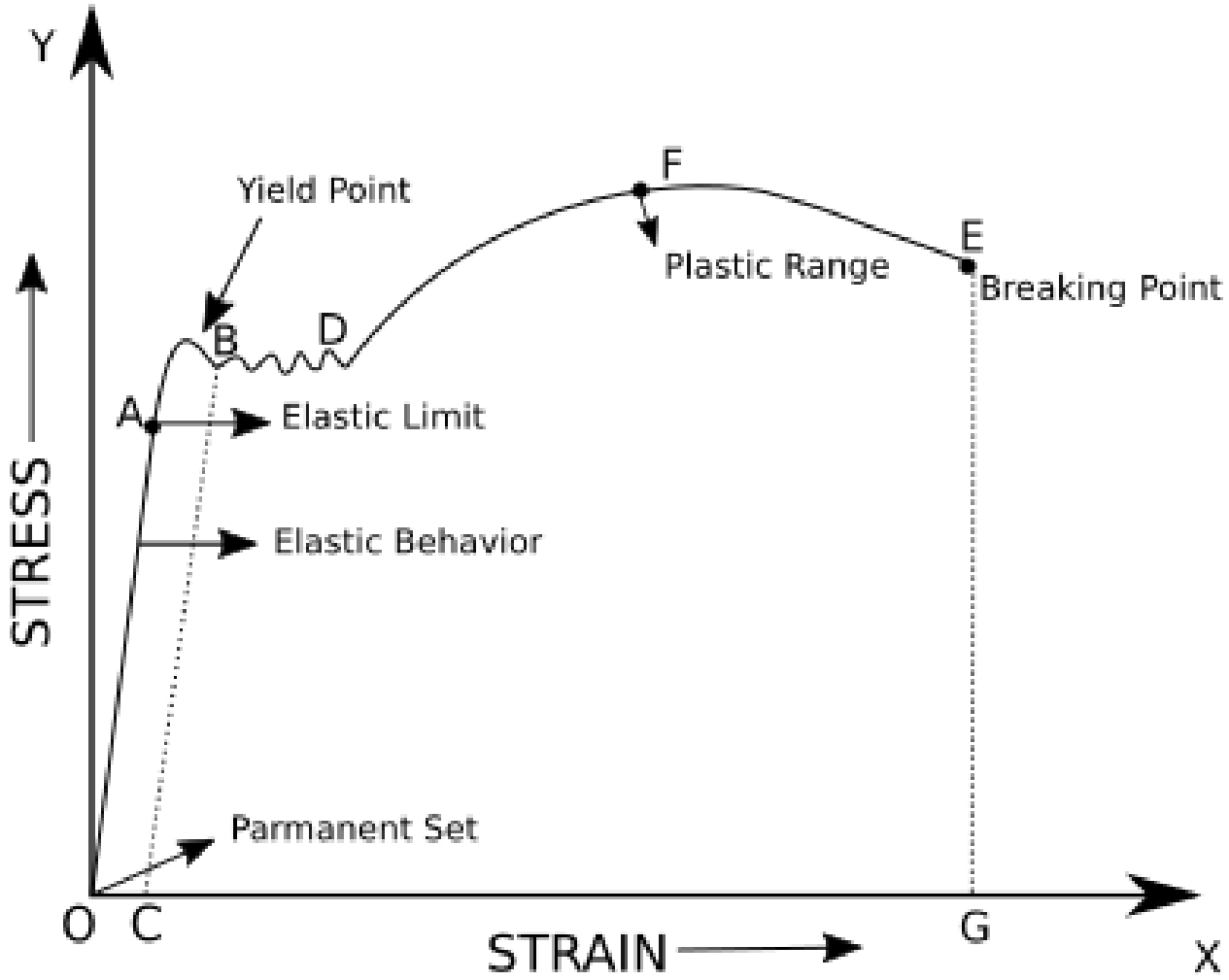
$$\text{stress} \propto \text{strain}$$

Or

$$\frac{\text{stress}}{\text{strain}} = E \qquad \text{(a constant)}$$

The constant of proportionality E is called the modulus of elasticity or coefficient of elasticity. This is the measure of elasticity of the material of the body.

Stress - Strain Relationship



A graph showing the variation of stress with strain is called stress-strain diagram.

- The straight and sloping part **OA** of the curve shows that the strain produced is directly proportional to the stress applied.
- Hooke's law is obeyed perfectly up to **A** and on the removal of the stress, it will recover its original condition of zero strain, represented by **O**.
- When the stress is continually increased in the case of a solid, a point is reached at which the strain increases rapidly.
- he stress at which the linear relationship between stress and strain ceases to hold good is referred to as the *elastic limit* of the material.
- Thus, if the material happens to be in the form of a bar or a wire under stretch, it will recover its original length on the removal of the stress so long as the stress is below the elastic limit, but if this limit is exceeded, it will fail to do so and will acquire what is called as *permanent set*.

- As soon as the elastic limit is crossed, the strain increases more rapidly than the stress, and the graph curves along **AB**, the extension of the wire now being partly elastic and partly plastic.
- Hence, on being unloaded here, say, at the point **B**, it does not come back to its original condition along **AO**, but takes the dotted path **BC**, so that there remains a residual strain **OC** in it, which is *permanent set* acquired by the wire.
- The strain left behind in the material even after the removal of the applied load is called *residual strain.*
- Beyond the point **B**, for practically little or no increase in stress (or the load applied) there is a large increase in strain (i.e., in the extension produced) up to **D**, so that the portion **BD** of the graph is an irregular wavy line, the stress corresponding to **D** being less than that corresponding to **B**.

- This point **B** where the large increase in strain commences is called the *yield point*, the stress corresponding to it being known as the *yielding stress*.
- The yielding ceases at **D** and the material now becomes plastic.
- Stress being mainly due to shear stress rather than simple tension, further extension can be produced gradually by increasing the load so that the portion **DF** of the graph is obtained.
- The cross-section of the wire decreases uniformly with extension (or strain) up to **F** and hence volume remaining constant.
- The maximum load (or force) to which the wire is subjected divided by its original cross sectional area is called the *ultimate strength* or the *tensile strength* of the wire and is also termed as *breaking stress*.

- The extension of the wire goes on increasing beyond **F** without any addition to the load, even if the load is reduced a little, and wire behaves as though it were literally *flowing down*.
- This is because of a faster rate of decrease of its cross-sectional area at some section of its length where local constriction, called a *neck* begins to develop.
- With this even if the load is not increased, the load per unit area or the stress becomes considerably greater there, bringing about corresponding increase in strain or extension in the wire.
- The load is therefore decreased, i.e., the stress reduced at this stage and the wire finally snaps or breaks at **E**, which thus represents the *breaking point* for it.

Elastic Fatigue:

- We recall that, if the maximum stress in the specimen does not exceed the elastic limit of the material, the specimen returns to its initial condition when the load is removed.
- Such a conclusion is correct for loading repeated a few dozen or even a few hundred times.
- However, it is not true when loading is repeated thousands or millions of times. In such cases rupture will occur at a stress much lower than the static breaking strength, this phenomenon is known as *fatigue*.
- A fatigue failure is of a brittle nature even for materials that are normally ductile.

- A body subjected to repeated strains beyond its elastic limit, has its elastic properties greatly impaired, and may break under a stress less than its normal breaking stress even within its elastic limit. This is called elastic fatigue.
- Fatigue must be considered in the design of all structural and machine components that are subjected to repeated or to fluctuating loads.
- The number of loading cycles that may be expected during the useful life of a component varies greatly. For example, a beam supporting an industrial crane may be loaded as many as two million times in 25 years (about 300 loading per working day) an automobile crankshaft will be loaded about half a billion times if the automobile is driven 320,000 km, and an individual turbine blade may be loaded several hundred billion times during its lifetime.

Strain Hardening:

- When a metal is stressed beyond its elastic limit, it enters the plastic region in which the residual strain remains upon unloading.
- When the load is further increased, the material hardens and becomes stronger, i.e it's more difficult to deform the material as the strain increases and hence it's called *strain hardening*.
- This tends to increase the strength of the material and decreases its ductility. It is region between *yield point* and *ultimate tensile strength*.

Strain softening:

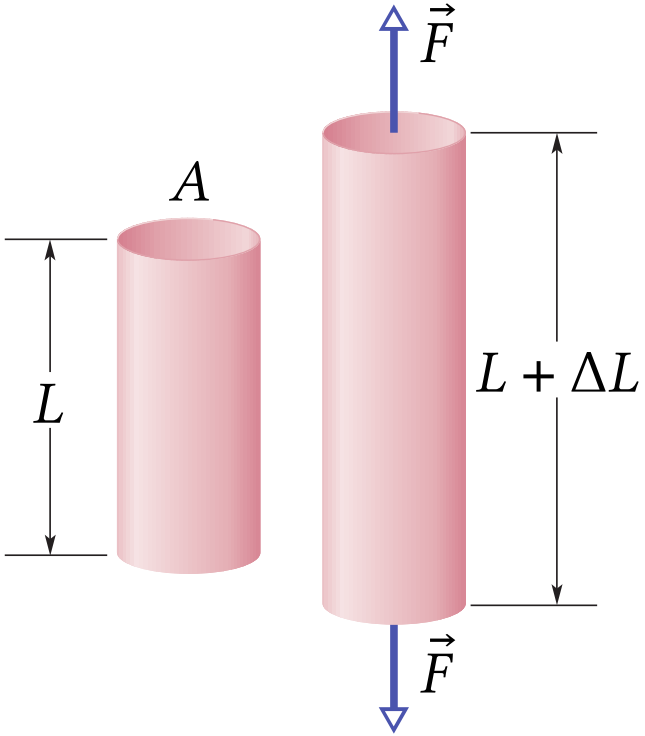
- Strain softening is defined as the region in which the stress in the material is decreasing with an increase in strain.
- It is observed after the yield point.
- This is due to brittleness and heterogeneity of the material.

Modulus of Elasticity

1. Linear Elasticity or Young's modulus (Y):

When the deforming force is applied to the body, along a particular direction, the change per unit length in that direction is called *longitudinal* or *linear* or *elongation strain*, and the force developed applied per unit area of cross section is called *longitudinal* or *linear stress*.

> The ratio of *longitudinal stress* to *linear strain* within elastic limit is called the *Young's modulus* and is denoted by Y.



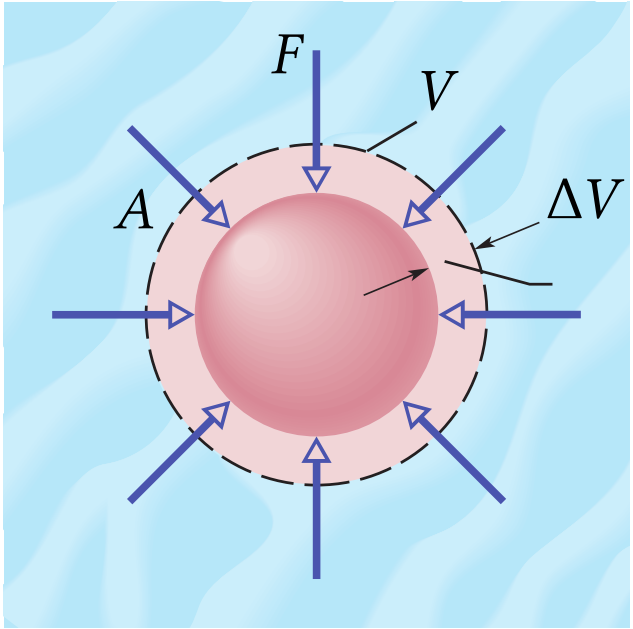
If F is the force applied normally to a cross-sectional area A, then the stress is F/A. If L is original length and ΔL is the change in length due to the applied force the strain is given by $\Delta L/L$ so that,

$$Y = \frac{\text{Normal stress}}{\text{Longitudinal strain}} = \frac{F/A}{\Delta L/L} = \frac{FL}{A\,\Delta L} \qquad (\text{in N/m}^2)$$

2. Bulk modulus (K):

When the deforming force is applied uniformly along the normal to the surface of a body, it produces a *volume strain* (without changing its shape in case of solid bodies). The applied force per unit area gives the *normal stress* or pressure.

> The ratio of the *normal stress* to the *volume strain* within elastic limit is called the *bulk modulus* and is denoted by K.

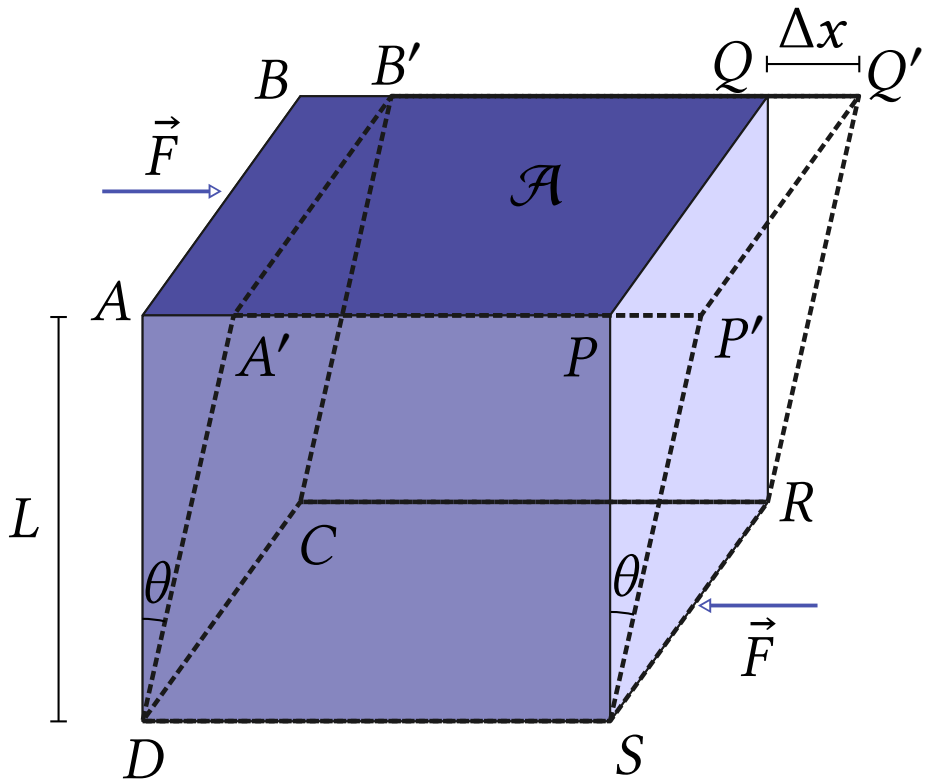


If F is the force applied uniformly and normally on a surface of area A, the stress or pressure is F/A and if ΔV is the change in volume produced in an original volume V, the strain is given by $\Delta V/V$ and therefore

$$K = \frac{\text{Normal stress}}{\text{Volume strain}} = \frac{F/A}{\Delta V/V} = \frac{FV}{A\,\Delta V}$$

3. Rigidity Modulus (η):

In this case, while there is a change in the shape of the body, there is no change in its volume. It takes place by the movement of layers of the body one over the other.



Consider a rectangular solid cube whose lower face $DCRS$ is fixed and to the upper face $ABQP$ a tangential force F is applied in the direction as shown. Under the action of this force, the layers of the cube which are parallel to the applied force slide one over the other such that point A shifts to A', B to B', P to P' and Q to Q'.

The body has turned through an angle. This angle is called the *angle of shear* or *shearing strain* θ.

$$\theta = \frac{PP'}{PS} = \frac{\Delta x}{L}$$

Tangential stress is equal to the force F divided by area A of the face $ABQP$.

$$\text{tangential stress} = \frac{F}{A}$$

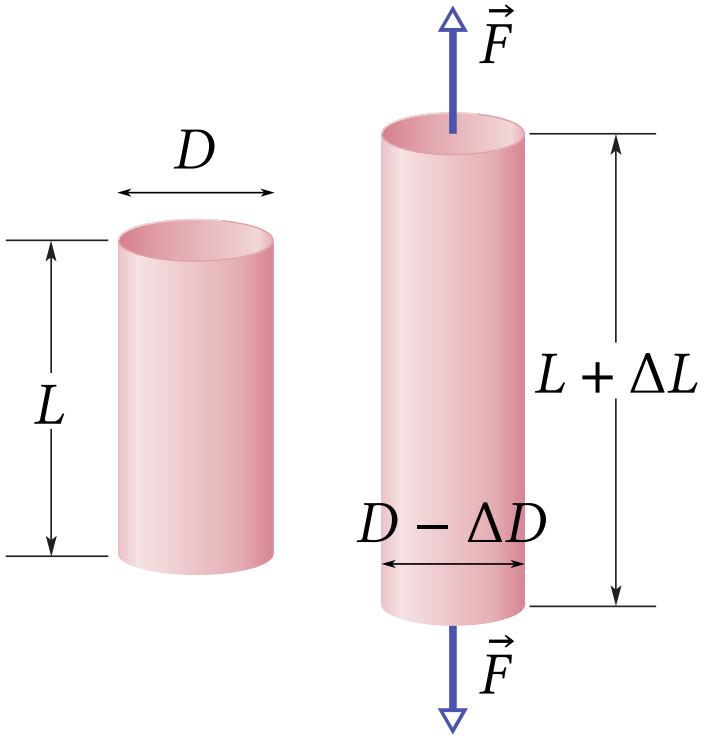
The *rigidity modulus* is defined as the ratio of the *tangential stress* to the *shearing strain* within elastic limit is called the *rigidity modulus* and is denoted by η.

$$\eta = \frac{\text{Tangential stress}}{\text{Shearing strain}} = \frac{F/A}{\Delta x/L} = \frac{FL}{A\,\Delta x}$$

Poisson's Ratio (σ)

In case of any deformation taking place along the length of a body like a wire, due to a deforming force, there is always some change in the thickness of the body. This change which occurs in a direction perpendicular to the direction along which the deforming force is acting is called *lateral change*.

> Within elastic limits of a body, the ratio of *lateral strain* to *longitudinal strain* is a constant and is called *Poisson's ratio*.

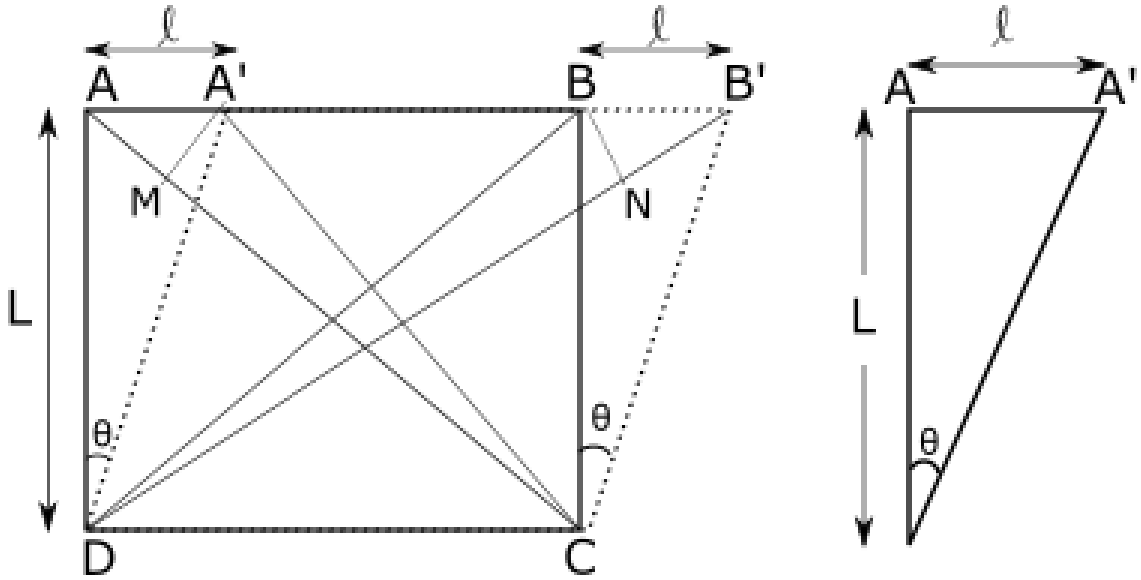


If a deforming force acting on a wire of length L produces a change in length ΔL accompanied by a change in diameter ΔD in it which has a original diameter of D, then the lateral strain is $= \Delta D/D$ and longitudinal strain is $= \Delta L/L$.

$$\therefore \quad \sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{\Delta D/D}{\Delta L/L} = \frac{L}{\Delta L}\frac{\Delta D}{D}$$

Poisson's ratio is a dimensionless quantity.

Equivalence of Shear Strain to Compression and Extension



Consider a cube of length L whose lower surface CD is fixed to a rigid support. Let $ABCD$ be its front face. If a tangential force F is applied on the upper surface along AB, it causes relative displacement at different parts of the cube, so that A moves to A' and B moves to B' through a small angle. Due to this the diagonal AC will be shortened to $A'C$ and diagonal DB will be increased in length to DB'.

As diagonal DB increases to DB' and diagonal AC is compressed to $A'C$, therefore

$$\text{extension strain along } DB \text{ is} = \frac{NB'}{DB}$$

$$\text{and extension strain along } AC \text{ is} = \frac{MA}{AC}$$

If L is the length of each side of the cube, then by Pythagoras theorem,

$$DB = AC = \sqrt{2}\,L$$

The shear is very small in actual practice and therefore triangles AMA' and BNB' are isosceles right angle triangles. From these right triangles we have, with $AA' = BB' = l$,

$$NB' = BB' \cos 45^\circ = l/\sqrt{2}$$

and

$$MA = AA' \cos 45^\circ = l/\sqrt{2}$$

Let $\theta = l/L$ be the angle of shear which is very small in magnitude. Therefore,

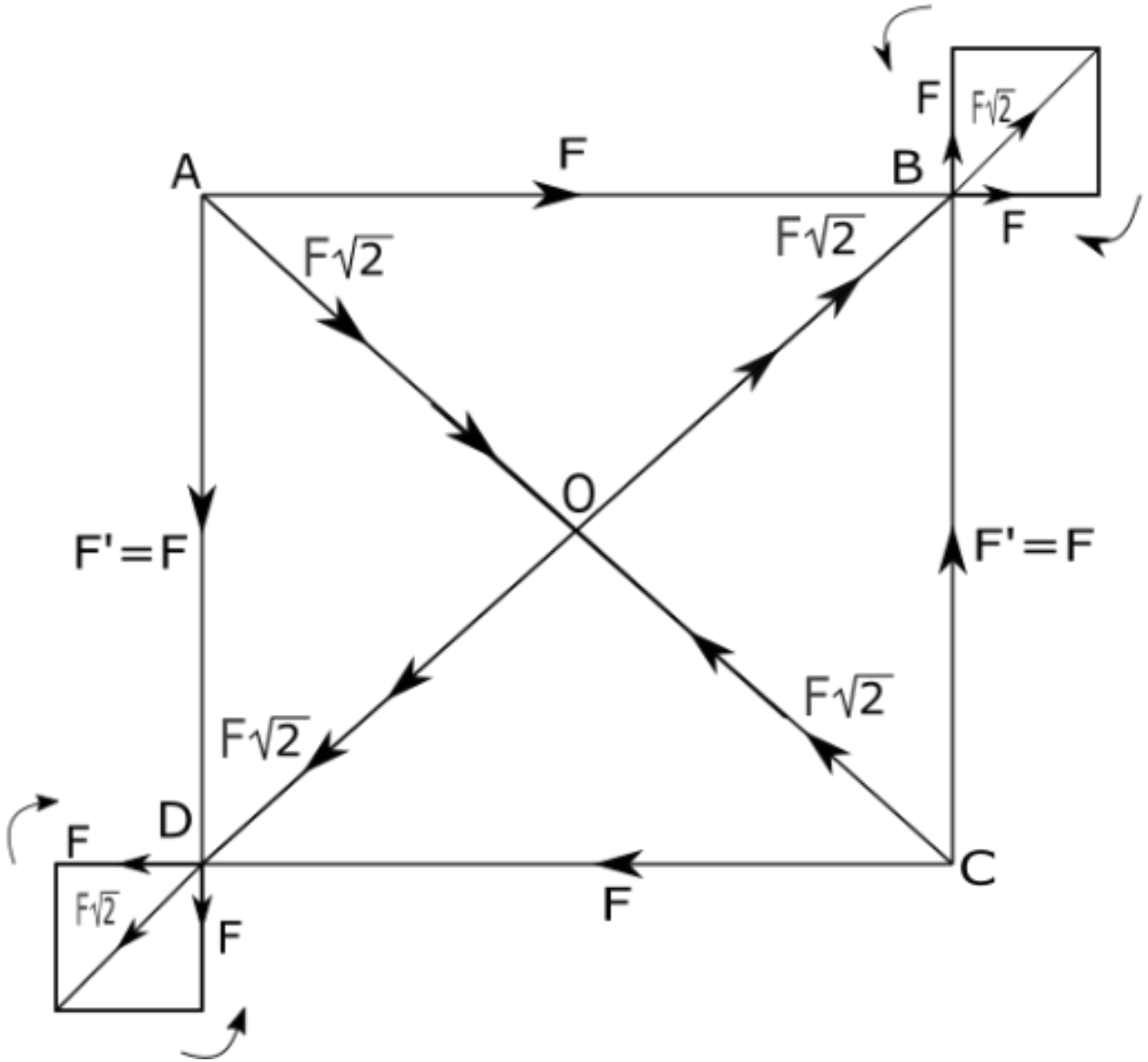
$$\text{extension strain along DB is} = \frac{NB'}{DB} = \frac{l/\sqrt{2}}{\sqrt{2}\,L} = \frac{l}{2L} = \frac{\theta}{2}$$

$$\text{and extension strain along } AC \text{ is} = \frac{MA}{AC} = \frac{l/\sqrt{2}}{\sqrt{2}\,L} = \frac{l}{2L} = \frac{\theta}{2}$$

From these two relations it is clear that a simple *shear strain* θ is equivalent to an *extension strain* and a *compression strain* at right angles to each other and each with value $\theta/2$.

The converse is also true. That is, simultaneous equal compression and extension at right angles to each other are equivalent to shear.

Equivalence of Shearing Stress to a Compressive Stress and a Tensile Stress



- *Compressive stress* (or compression) is the stress state caused by an applied load that acts to reduce the length of the material in the axis of the applied load.
- *Tensile stress* is the stress state caused by an applied load that tends to elongate the material in the axis of the applied load, in other words the stress caused by pulling the material.
- In the case of the cube, if F was the only force acting on its upper face, it would move the body in the direction of this force.
- Since, however, the cube is fixed at its lower face DC, an equal and opposite force comes into play in the plane of this face, giving rise to a couple FL (L is the length of each edge of the cube and hence the perpendicular distance between the two faces F and F'), tending to rotate the cube in clockwise direction.
- Again, since the cube does not rotate, it is obvious that the plane DC applies an equal and opposite couple $F'L$, by exerting forces along the faces AD and CB tending to rotate it in the anticlockwise direction as shown.

- Thus, because the cube is in equilibrium under the two couples, we have

$$FL = F'L \qquad \text{or} \qquad F = F'$$

  i.e., a tangential force F applied to the face AB results in an equal tangential force acting along all the other faces of the cube in the directions shown in the figure.

- The resultant of the two forces F and F' along AB and CB respectively is $F\sqrt{2}$ along OB, and of those acting along AD and CD is also $F\sqrt{2}$ along OD.

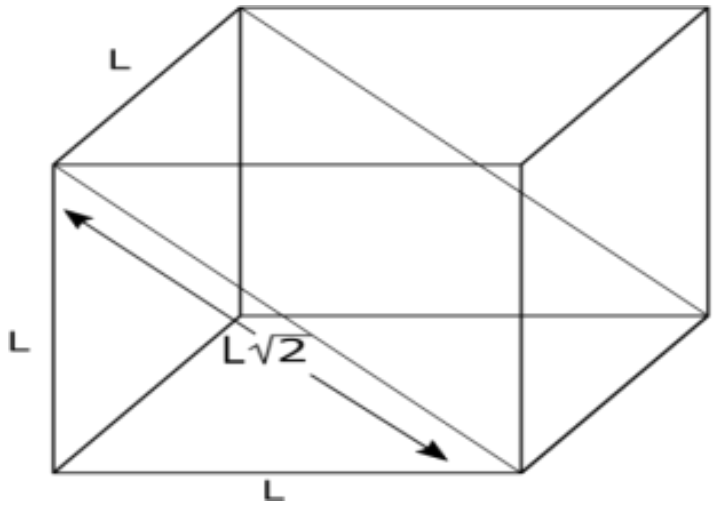
- Thus, an outward pull acts on the diagonal DB of the cube at B and D, resulting in its extension.

- Similarly, an inward pull acts on the diagonal AC at A and C, thereby bringing about its compression.

- Thus, a tangential force F applied to one face of a cube gives rise to a force $F\sqrt{2}$ outward along one diagonal BD and an equal force $F\sqrt{2}$ inward along the other diagonal AC of the cube, resulting in an extension of the former and compression of the latter.

- Now, if the cube be cut up into two halves, by a plane passing through AC and perpendicular to the plane of the paper, each face, parallel to the plane, will have an area $L \times L\sqrt{2} = L^2\sqrt{2}$ and clearly the outward force $F\sqrt{2}$ along BD will be acting perpendicularly to it. Then we have,

$$\text{Linear tensile stress along } BD = \frac{F\sqrt{2}}{L^2\sqrt{2}} = \frac{F}{L^2}$$



- Similarly, if we cut the cube into two halves, by a plane passing through BD and perpendicular to the plane of the paper, we shall have an inward force $F\sqrt{2}$ along AC acting perpendicularly to a face of an area $L^2\sqrt{2}$. Then we have,

$$\text{Linear compressive stress along } AC = \frac{F\sqrt{2}}{L^2\sqrt{2}} = \frac{F}{L^2}$$

- As learned earlier, F/L^2 is the shearing stress over the face AB of the cube, which produces the shear θ in it.
- Thus, it is clear that a *shearing stress* is equivalent to an equal *linear tensile stress* and an equal *linear compressive stress* at right angles to each other.

Relations Between the Moduli of Elasticity

Let α be the elongation per unit length (longitudinal strain) per unit stress along the direction of the forces.

Let β be the contraction per unit length (lateral strain) per unit stress in a direction perpendicular to the forces.

1. Young's modulus Y:

$$Y = \frac{\text{normal stress}}{\text{longitudinal strain}} = \frac{1}{\frac{\text{longitudinal strain}}{\text{normal stress}}} = \frac{1}{\alpha}$$

2. Bulk modulus K:

$$K = \frac{1}{3(\alpha - 2\beta)}$$

3. Rigidity modulus η:

$$\eta = \frac{1}{2(\alpha + \beta)}$$

4. Poisson's Ratio σ:

$$\sigma = \frac{\text{lateral strain}}{\text{longitudinal strain}} = \frac{\frac{\text{lateral strain}}{\text{stress}}}{\frac{\text{longitudinal strain}}{\text{stress}}} = \frac{\beta}{\alpha}$$

1. Relations between Young's modulus Y and bulk modulus K:

$$K = \frac{1}{3(\alpha - 2\beta)} = \frac{1}{3\alpha(1 - 2\beta/\alpha)} = \frac{1/\alpha}{3(1 - 2\beta/\alpha)}$$

With $1/\alpha = Y$ and $\beta/\alpha = \sigma$ we can write,

$$K = \frac{Y}{3(1 - 2\sigma)}$$

2. Relations between Young's modulus Y and rigidity modulus η:

$$\eta = \frac{1}{2(\alpha + \beta)} = \frac{1}{2\alpha(1 + \beta/\alpha)} = \frac{1/\alpha}{2(1 + \beta/\alpha)}$$

With $1/\alpha = Y$ and $\beta/\alpha = \sigma$ we can write,

$$\eta = \frac{Y}{2(1 + \sigma)}$$

3. Relations between Young's modulus Y, bulk modulus K and rigidity modulus η:

We had,

$$\alpha - 2\beta = \frac{1}{3K}$$

$$\text{and } 2\alpha + 2\beta = \frac{1}{\eta}$$

Adding one another and eliminating β gives,

$$3\alpha = \frac{1}{3K} + \frac{1}{\eta}$$

Writing $\alpha = 1/Y$ we get,

$$\frac{1}{Y} = \frac{1}{9K} + \frac{1}{3\eta}$$

Or

$$Y = \frac{9\eta K}{3K + \eta}$$

4. Relation between bulk modulus K, Rigidity modulus η and Poisson's ratio σ:

From equations,

$$K = \frac{Y}{3(1-2\sigma)} \qquad \text{and} \qquad \eta = \frac{Y}{2(1+\sigma)}$$

eliminating Y we get,

$$3K(1-2\sigma) = 2\eta(1+\sigma)$$

$$\implies \quad 3K - 2\eta = \sigma(2\eta + 6K)$$

Or

$$\sigma = \frac{3K - 2\eta}{2\eta + 6K}$$

Limiting Values of Poisson Ratio

From equations,

$$K = \frac{Y}{3(1-2\sigma)} \qquad \text{and} \qquad \eta = \frac{Y}{2(1+\sigma)}$$

eliminating Y we get,

$$3K(1-2\sigma) = 2\eta(1+\sigma)$$

Here K and η are positive quantities. We can see two cases here,

i) If σ is positive then RHS will be positive. Then for equality to hold LHS also should be positive. For this it has to be,

$$1-2\sigma > 0 \qquad \text{or} \qquad \sigma < 0.5$$

ii) If σ is negative then LHS will be positive. Then for equality to hold RHS also should be positive. For this it has to be,

$$1+\sigma > 0 \qquad \text{or} \qquad \sigma > -1$$

- Thus the limiting values of σ are -1 and 0.5.
- A negative value of σ means that a body on being extended should expand laterally and one can hardly expect this to happen (we know of no substance so far).
- Similarly, a value of $\sigma = 0.5$ would mean that the substance is perfectly incompressible, and we do not know of any such substance either.
- In actual practice, the value of σ is found to lie between 0.2 and 0.4.

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